Two types of enzymes are required for digestion of carbohydrates – Amylases and Disaccharidases. Amylases are present in saliva and pancreatic juice, while disaccharidases are present in the brush border membrane of the intestine (Figure 2.11).

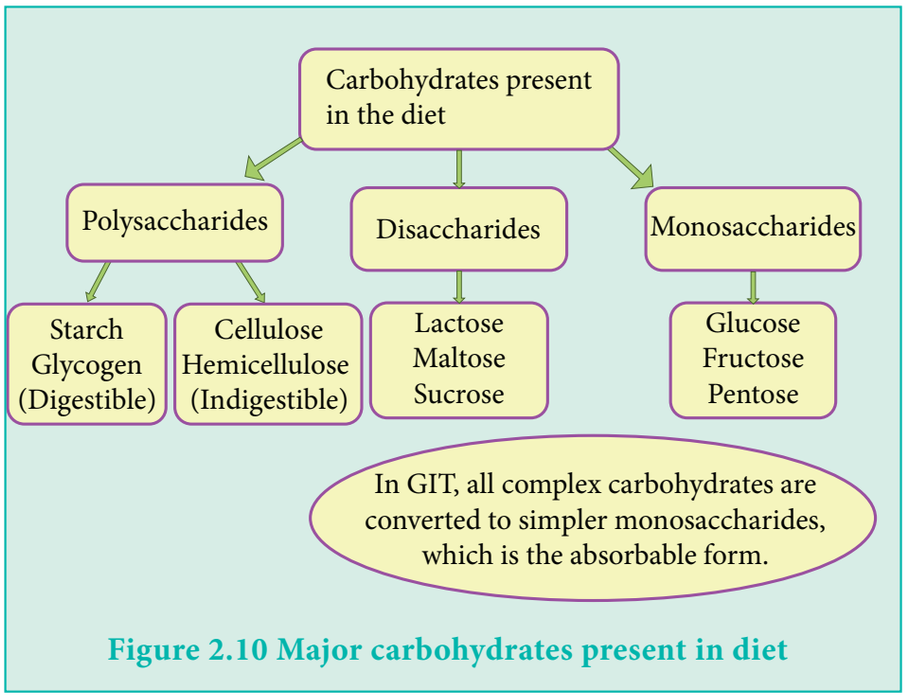


Figure 2.10 Major carbohydrates present in diet

Digestion in mouth

Simpler carbohydrates and carbohydrates present in milk and juices escape digestion in mouth. The digestion of the digestible polysaccharides like starch and glycogen begins in mouth in humans by action of the salivary enzyme, Ptyalin, also called as α-amylase. This enzyme with an optimum pH of 6.8, hydrolyzes the α1→4 glycosidic bonds to release smaller oligosaccharide fragments and a highly branched limit dextrin (as the inner α1→4 linkages are inaccessible to the salivary amylase) (Figure 2.12). However, the digestion of carbohydrates by salivary amylase is limited because of its lower residual time in the mouth and oesophagus. As soon as the bolus reaches the stomach, the salivary amylase is inactivated because of the prevailing low pH.

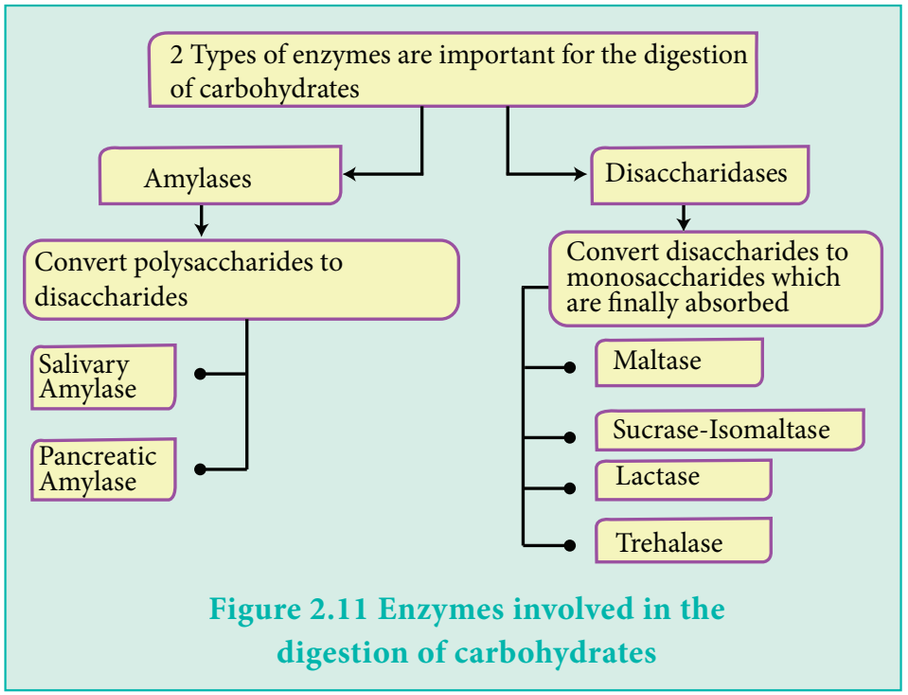


Figure 2.11 Enzymes involved in the digestion of carbohydrates

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